



Smart University Data Engineering Platform

Batch No: 6

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Abstract

Student Services: Provides Face Recognition Attendance, Internal Marks Notifications, and Free Classroom Finder by collecting and processing student, marks, and timetable data.

Faculty Services: Provides Automated Question Paper Generation using handouts/PPTs and Instant Classroom/Lab Replacement using processed timetable and room-availability data.

Campus Services: Provides Smart Bus Tracking with Alternative Bus Suggestions and Canteen Digital Token with Live Queue Management by processing transport and order-related data.

Data Engineering Process: Uses Data Ingestion to collect data from different sources, Data Cleaning to remove incorrect or incomplete data, Data Transformation to convert it into useful formats, and Data Storage to maintain the processed data for the university services.

Data-Driven Services: The stored and processed data is used to identify students through attendance, notify students when marks are published, find available rooms, generate question papers, replace classrooms, track buses and delays, and manage canteen orders and tokens.

Introduction

University Operations: Students, faculty, classrooms, transport, and canteen services involve many routine activities that require continuous data handling.

Existing Challenges: Attendance marking, marks communication, finding vacant rooms, preparing question papers, replacing unavailable classrooms, tracking buses, and managing canteen queues often require manual coordination.

Need for a Centralized System: These activities involve different types of data such as student records, academic content, timetables, room availability, bus routes, and food orders.

Our Approach: The project connects these data sources and processes them to provide **seven practical university services** — attendance, marks notification, classroom finding, question-paper generation, classroom replacement, bus tracking, and canteen queue management.

FDE Perspective: The project demonstrates how **data is collected, cleaned, transformed, validated, stored, and used** to support real-world university operations.

Tools & Technologies

Data Generation & Ingestion — Python + Pandas

Generate and ingest structured university datasets such as student, academic, timetable, transport, and canteen data.

Streaming Ingestion — Apache Kafka

Designed to handle real-time events such as attendance, bus updates, canteen orders, and notifications.

Large-Scale Data Processing — Apache Spark

Designed to clean, transform, aggregate, and process large university datasets as the system scales.

Data Storage — PostgreSQL + MinIO

Store structured operational data and raw/processed datasets.

Backend / APIs — FastAPI

Provide services for attendance, marks, classroom availability, question-paper generation, room replacement, bus tracking, and canteen management.

AI / Document Processing — Python + AI Models

Support face recognition and question-paper generation from handouts/PPTs.

Frontend — React.js

Provide interfaces for students, faculty, and campus operations.

Development — VS Code + Git/GitHub

Development, testing, collaboration, and version control.

Literature Survey – Existing Approaches

#	Paper Link	Author(s)	Drawback / Research Gap	Future Work
1	A Systematic Review of the IoT in Smart University: Model and Contribution	Senny Luckyardi, Ratih Hurriyati, Disman & Puspo Dewi Dirgantari	Existing IoT models for smart universities are incoherent and lack strong theoretical frameworks specific to university environments.	Develop integrated IoT adoption models and validate them in real smart university environments. (Ejournal UPI)
2	Internet of Things and Its Applications to Smart Campus	Nadire Cavus, Seipati E. Mrwebi, Imran Ibrahim, Temiloluwa Modupeola & Albert Y. Reeves	Findings have limited generalizability because the review is based on selected studies and databases.	Test proposed approaches through real-world smart campus implementations and broader empirical studies. (Online Journals)
3	Guidance in Designing a Smart Campus: A Systematic Literature Review	Hadad Al Akbar, Muhammad Reza Faturrahman, Sidharta et al.	Provides general design guidance but lacks sufficient practical validation of a complete smart campus architecture.	Develop and evaluate integrated smart campus frameworks using emerging technologies such as IoT, AI, blockchain and edge computing. (ScienceDirect)
4	Paper focuses on learning analytics benefits and challenges	S. El Alfy, J. Marx Gómez & A. Dani	Learning analytics adoption faces challenges related to privacy, ethics, data quality and institutional implementation.	Investigate practical frameworks for ethical and effective implementation of learning analytics in universities.
5	Learning Analytics Methods, Benefits, and Challenges in Higher Education	Sandra G. Nunn, John T. Avella, Therese Kanai & Mansureh Kebritchi	The field lacks clearly established requirements and consistent approaches for implementing learning analytics across institutions.	Develop clearer implementation guidelines and evaluation frameworks for learning analytics systems. (ERIC)

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6	A Decade of Research into Big Data and Analytics in Higher Education	Ana Stojanov & Ben Kei Daniel	Research mainly focuses on learning, with limited use in teaching and administration.	Explore broader applications across higher education.
7	Big Data Analytics in Higher Education: A Systematic Review	Zeynep Aytaç & Hasan Şakir Bilge	Lack of unified frameworks across stakeholders and learning environments.	Develop comprehensive analytics models for stakeholders.
8	Utilising Learning Analytics to Support Study Success in Higher Education	Dirk Ifenthaler & Jane Y.-K. Yau	Limited evidence that interventions consistently improve student success.	Conduct longitudinal intervention studies to measure impact.
9	The Efficacy of Learning Analytics Interventions in Higher Education	A. L. Sønderlund et al.	Interventions show mixed effectiveness and inconsistent evaluation methods.	Standardize evaluation methods and conduct stronger experiments.
10	Educational Data Mining and Learning Analytics: An Updated Survey	Cristóbal Romero & Sebastián Ventura	Overlapping concepts make standardization and integration challenging	Develop integrated and explainable data-driven systems.

Objectives

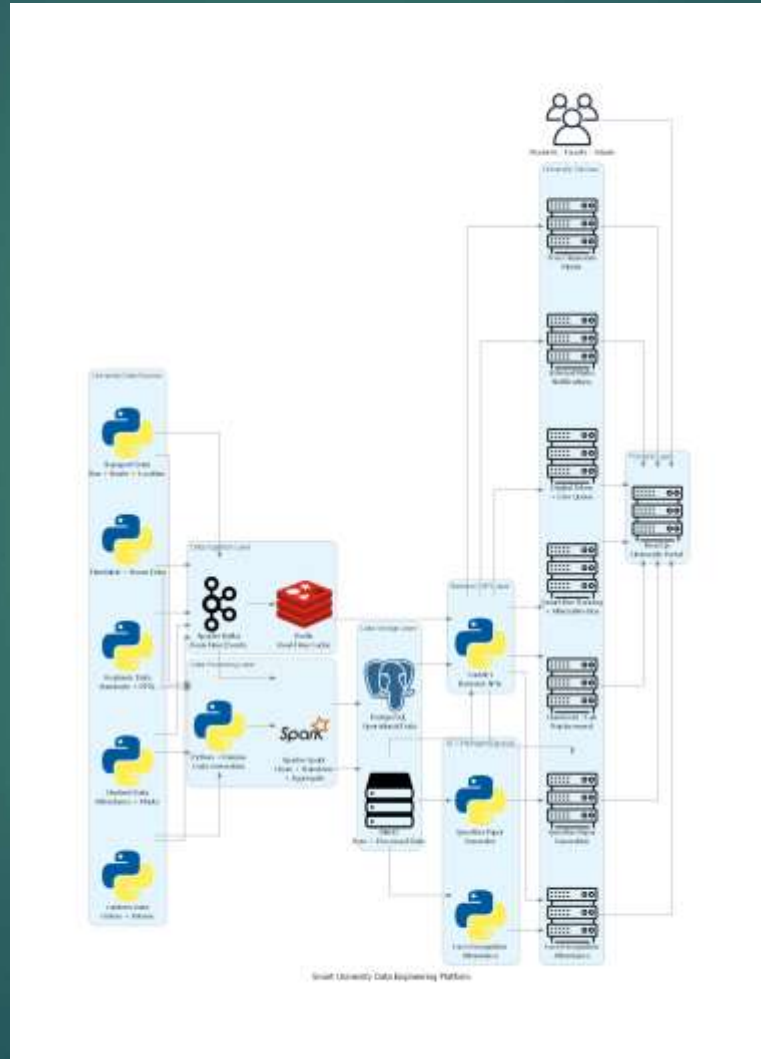
Improve Student Services: Provide automated **Face Recognition Attendance, Internal Marks Notifications, and Free Classroom Finder** for faster and easier student access.

Support Faculty Activities: Automate **Question Paper Generation** using academic materials and provide **Instant Classroom/Lab Replacement** when an assigned room becomes unavailable.

Improve Campus Operations: Provide **Smart Bus Tracking with Alternative Bus Suggestions** and **Canteen Digital Tokens with Live Queue Management**.

Centralize University Data: Organize and process data from **students, academics, timetables, classrooms, transport, and canteen operations** through a common data-driven platform.

Enable Data-Driven University Services: Convert processed university data into **real-time information, automated actions, notifications, availability results, and operational decisions** that reduce manual work and improve efficiency.



Methodology, System Design, and Architecture



Student Services Architecture:

Face Recognition Attendance identifies registered students and records attendance; Internal Marks Notification processes published marks and sends student notifications; Free Classroom Finder analyzes timetable and room data to identify available classrooms.

Faculty Services Architecture:

Question Paper Generator processes selected handouts/PPTs and generates structured question papers; Classroom/Lab Replacement checks room availability and assigns a suitable replacement when the original room is unavailable.

Transport Service Architecture:

Smart Bus Tracking processes route, stop, location, and timing data to display bus status, while delay information is used to provide alternative bus suggestions.

Canteen Service Architecture:

Digital Token and Live Queue processes food orders, generates a token and receipt, maintains the order through Ordered → Waiting → Processing → Ready → Collected, and updates the student and canteen display.

Common System Architecture:

All features follow a common flow of **Data/Input** → **Processing** → **Database** → **Backend/API** → **Feature Interface** → **User Output**, allowing student, faculty, and campus services to use processed university data.

Use of Modern Tools and Technologies

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Implementation, Coding Quality, and Functionality

Modular Implementation:

Each of the seven features is implemented as a separate functional module with reusable components and APIs.

Data Processing:

Data is collected, validated, cleaned, transformed, and stored before being used by the respective features.

Feature Functionality:

The system supports attendance, marks notifications, classroom finding, question-paper generation, room replacement, bus tracking, and canteen queue management.

Coding Quality:

The application follows a modular and organized structure with separate feature logic, database operations, and backend services for easier maintenance.

Testing & Validation:

Complete workflows are tested to verify correct data processing, database updates, notifications, status changes, and final outputs.

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